The Transportation Network Design Problem (TNDP) focuses on choosing the best set of road projects from several options to get the most benefit for the cost. Many methods have been use problem over the years.This study checks how well PSO works for a large city network, using Dhaka, Bangladesh as a case study. The experiment used estimated data from the city's current roa proposed new projects.

A PSO based Transportation Network Design Optimization of the Mega City Dhaka

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***Abstract*— Transportation network is a key issue for urban areas and is require to update to fulfill the growing demand modifying existing road(s) and/or constructing new road(s) considering various constraints. In general, transportation network design problem (TNDP) concerns optimal selection of several projects from various alternative proposed projects (with estimated costs) maintaining associated constraints to ensure benefit versus cost. Various approaches have been investigated to solve TNDP in last several decades. Recently, particle swarm optimization (PSO) based method is shown to outperform other methods. However, the most of the existing methods, including PSO, tested on the small scale TNDP. Therefore, the aim of this study is to identify the effectiveness of the PSO based method for a mega city heaving large number nodes and arcs. The mega city considered in this study is Dhaka, the capital of Bangladesh. The experiment has been conducted on the roughly estimated data for exiting network as well as proposed projects. Experimental results for revealed that the method able to select projects for optimal (or near optimal) utilization of a given budget amount. The selected projects are seem to be effective to increase traffic flow while observe on the schematic diagram of the road network.**

Keywords*— *Transportation network design; Particle swarm optimization component;

I. Introduction

A transportation network is a network of roads, streets which permits vehicular movement. It is a key element of society and its quality depends on easy vehicular movement [1]. A network is require to update to fulfill the growing demand modifying existing road(s) and/or constructing new road(s) considering some constraints. Transportation network development is always bears a challenging issue to maximize user benefit through optimal utilization of resources. In general, transportation network design problem (TNDP) concerns optimal selection of several projects from various alternative proposed projects (with estimated costs) maintaining associated constraints to ensure benefit versus cost. Moreover, the task of TNDP becomes more complex when network size increases. Therefore, TNDP has been emerged as a challenging optimization task.

Various approaches have been taken to solve TNDP in last several decades [2-5]. A brief survey of earlier methods is available in [6]. Recently, a number swarm intelligence based methods are studied to solve TNDP and among them particle swarm optimization (PSO) based method is the most recent one [2, 3]. The PSO based method is shown to outperform the methods based on ant colony optimization (ACO) and hybrid methods [4]. However, the methods tested on the small scale TNDP such as Sioux Falls network with only 24 nodes. Therefore, the aim of this study is to identify the effectiveness of the PSO based method for a mega city having large number of nodes and arcs. The mega city considered in this study is Dhaka, the capital of Bangladesh.

The rest of this paper is organized as follows. Section II first describes TNDP as an optimization task and then explains the PSO based method to solve TNDP. Section III investigates PSO based TNDP solutions for the mega city Dhaka. The section includes brief description of Dhaka city and data preparation. Finally, the paper concludes with a few remarks in Section IV.

II. TNDP and PSO based Method to Solve It

Transportation planning and development is a key issue for sustainable development of urban areas designing new transportation network(s) or redesigning (developing or modifying) existing ones. Its aim is to improve the mobility and relieve the traffic congestion [1]. In general there are two basic steps to update a network: determine the flow of vehicles through the network and determine appropriate of tasks owing to achieve optimal improvement. Appropriate tasks selection is a complex optimization issue to select which road(s) will be updated and/or where new road(s) will constructed within available budget considering a number of proposals or projects. TNDP may formulate in the following manner.

$G(N, A)$: The graph under consideration representing network with the set of nodes N and the set of arcs (representing roads), A; a representing an arc, $a \epsilon A$.

P : The set of origin(r) to destination(s) pair(OD), $P \subseteq \{(r, s) \in N \times N : r \neq s\}$.

d^{rs} : Travel demand for (OD) pair, where$(r, s) \in P$, assumed to be constant.

$\overline{A}$: The set of project arcs, $(\overline{A} \neq A)$.

y : Vector of $(y_a)_{a \epsilon \overline{A}}$, where $y_a$is binary decision.

$A(y)$: The set of arcs with selected project arcs, $A(y) = A \cup \{a \in \overline{A} : y_a = 1\}$.

$G(y)$: Graph G with selected project arcs.

Eq. (1): decides which roads to build or upgrade (city planner's decis
Eq. (2): determines how people will use those roads (drivers' behavic
This two-layer setup is called a bi-level optimization problem:
Upper level = government/planner decision
Lower level = drivers' route choice

$K_{rs}(y)$: The set of routes joining origin(r) to destination(s) in $G(y)$.
f_k : The flow of path $k \in K_{rs}(y)$.
δ_{ak} : 1 if arc $a \in A(y)$ lies on path k and otherwise 0.
B : Budget.
c_a : Construction cost of link $a \in \overline{A}$.
x_a : Number of traffic through link $a \in A(y)$.
$t_a(x_a)$: Volume-delay function of link $a \in A(y)$.

The aim of TNDP is to minimize total travel time (*T(y)*) satisfying the construction cost under budget (*B*) can be illustrated with the following equation (i.e., Eq. (1)).

$$\underset{y}{\text{Min}}\ T(y) = \sum_{a \in A(y)} x_a\, t_a(x_a)$$

$$\text{s.t.:} \sum_{a \in \overline{A}} c_a\, y_a \le B$$

$$y_a = 0 \text{ or } 1 \qquad \forall a \in A \tag{1}$$

In Eq. (1), $x(a)_{a \in A(y)}$ is the user equilibrium flow in $G(N, A(y))$ and found through the traffic assignment model using (2) for a given *y*.

$$\text{Min} \sum_{a \in A(y)} \int_0^{x_a} t_a(u)\, du$$

$$\text{s.t.: } f_k \ge 0 \quad \forall k \in K_{rs}(y) \quad \forall (r,s) \in P$$

$$\sum_{k \in K_{rs}(y)} f_k = d^{rs} \qquad \forall (r,s) \in P$$

$$x_a = \sum_{(r,s) \in P} \sum_{k \in K_{rs}(y)} f_k\, \delta_{ak} \quad \forall a \in A(y) \tag{2}$$

Considering Eq. (1) and Eq. (2) TNDP is a bi-level programing problem where (1 (upper level problem) seeks a decision vector *y* for minimizing the total travel time *T(y)* of the (assigned) traveller, and Eq. (2) (lower level problem) is the traffic assignment model which estimates the traveller flows, given the decision *y*.

Considering TNDP as a complex optimization task, a number of swarm intelligence based methods have been investigated in the last few years and PSO is the most recent one [7]. In short, PSO is a population based optimization method based on the simulation of the social behaviour of birds, bees or a school of fishes. PSO optimizes a problem by having a population of candidate solutions, here dubbed particles, and moving these particles around in the search-space according to simple mathematical formulae over the particle's position and velocity. Each particle's movement is influenced by its local best known position (P_i) and is also guided toward the global best (G), the best known positions in the search-space. The P_i and G are updated as better positions are found by particles. Formally, each particle in PSO changes position $X_i = \{x_{i1}, x_{i2}, x_{i3} \dots, x_{iD}\}$ based on its calculated velocity $V_i = (v_{i1}, v_{i2}, \cdots, v_{iD})$ that depends on its previous best position, $P_i = (p_{i1}, p_{i2} \dots p_{iD})$ and the best one among all the particles in the population $G = (g_1, g_2 \dots g_D)$. In each of iteration, each particle calculates its velocity using Eq. (3) and updates position using Eq. (4). Solving TNDP with PSO, each particle represents a set of feasible projects to optimize Eq. (1) and Eq. (2).

$$V_i^{(t)} = \omega\, V_i^{(t-1)} + c_1 * r_1 \left(P_i - X_i^{(t-1)}\right) + c_2 * r_2 \left(G - X_i^{(t-1)}\right), \tag{3}$$

$$X_i^{(t)} = X_i^{(t-1)} + V_i^{(t)}. \tag{4}$$

In Eq. (3), ω is inertia factor, c_1 and c_2 are learning factors, and r_1 and r_2 are vectors of random values between (0,1).

PSO require special considerations to handle TNDP because it was basically developed for continuous optimization problems. On the other hand, TNDP is a combinatorial optimization problem to identify appropriate vector *y* for $\overline{A}$ set of project arcs to be implemented with optimal utilization of budget. The TNDP-PSO method follows transformation between the one-dimensional real-valued space to the $|\bar{A}|$-dimensional binary to adapt PSO operation for TNDP. The particle position(X_i) as well as P_i, G and V_i in Eq. (3) and Eq. (4) are real numbers. But, to map a particle for $\overline{A}$ set of project arcs, the method considers X_i as its nearest integer in $[0, 2^{|\bar{A}|} - 1]$, and then transforms resulting integer into the base-2 number system as an $|\bar{A}|$-bit binary code. Each point in the binary code represents a project and 1 in this indicates the particular project is considered by the particle.

The network $G(N, A)$ is updated with projects of a decision vector y given by a particle and then the fitness (i.e., total travel time) of the updated network (i.e., $G(N, A \cup \overline{A})$ with new arcs of $\overline{A}$) is measured. Suppose, the number of traffic x are willing to go from node r to node s. This number x will be distributed among one or more routes and the total number of traffic x_a can be found, which want to move through an arc a. The empirical travel time function of an arc a for traffic flow x_a is:

$$t_a(x_a) = \alpha_a + \beta_a x_a^4, \tag{5}$$

where α is free flow travel time in hours and β is the parameter related to practical capacity in hours per thousand vehicles.

In the PSO, each particle *i* searches the solution X_i in the problem space with a range [0, X_{max}] (any other range can be translated to this range). In order to guide the particles effectively in the search space, the maximum moving distance during any iteration is clamped in between the maximum range.

To solve TNDP with PSO, the steps of TNDP-PSO algorithm are shown in Fig. 1. In initialization (Step 1), it defines number of particles and assigns a random solution X_i in (0, X_{max}) and a random velocity to each of the particle. Fitness of each particle is measured using Eq. (2), Eq. (5) and Eq. (1). Since total cost of projects selected by a solution must be within budget (Eq.

Step 1: Initialization.

Step 2: For each particle X_i in the swarm

a. Calculate velocity $V_i^{(t)}$ using Eq. (3)

b. Update Solution $X_i^{(t)}$ using Eq. (4)

c. Evaluate the Solution $X_i^{(t)}$ using Eq. (2), Eq. (5) and Eq. (1). Set fitness M if cost exceeds budget for the selected projects.

d. Update P_i if the new solution $X_i^{(t)}$ is superior to P_i

e. Update G if the new solution $X_i^{(t)}$ is superior to G

Step 3: Loop to **Step 2** until a termination criterion is met.

Step 4: Take the projects represents the global best solution G as an outcome.

Fig. 1. TNDP-PSO Algorithm.

(1)), an extremely large (i.e., worse) fitness value, say *M,* is assigned for the solution when it exceeds the budget. At this initial stage, previous best solution of each particle (P_i) is considered as the current solution of it and global best solution (G) is the best solution among them. TNDP-PSO checks termination criterion at the end of each iteration (in Step 2); usually a sufficiently good fitness of G or a maximum number of iterations is considered as the termination criteria. If a termination criterion does not meet, it continues updating positions of the particles again as indicates the loop to Step 2 from Step 3.

III. TNDP-PSO Solution for the Megacity Dhaka

This section gives brief overview of the megacity Dhaka, data preparation of it, and finally presents outcome of TNDP-PSO for the city.

A. The Dhaka City

Dhaka, the capital of Bangladesh and the nation's gateway as well as the economic, business, political, administrative, social, cultural hub of the country. It has now been turned into the 26th mega city and 10th most populous city in the world [8]. The city experiences rapid growth of many scattered and unplanned developments. The city transport system mainly depends on road-based and has facing an extreme levels of transportation problem. Moreover, population of Dhaka is increasing day by day and creating pressure on the existing transportation system. But transportation network development of the city is a critical issue due to limited spaces and funding problem.

Figure 2 (at the end of the paper) shows the schematic diagram of Dhaka city road network that is developed with help of Google map [9]. The network has 77 nodes and 192 arcs; and may consider in three clusters. The network covers Tongi (north most point and starting node as 1), Banasree (East most point, 17), Jatrabari (south most point, 32) and Gabtoli (west most point, 77). It is notable from the figure that most of the roads are North-South (South-North). Moreover, there are only few numbers of inter-links between the roads that are alternative for the movements. According to Google map the distance between Jatrabari to Tongi is 24.8 km and travel time is around 50 minutes without traffic. But in practical it requires several hours in working days due to huge traffic jam. At present, the traffic jam is the most critical issue for Dhaka. Network development of the city is an important issue of the country but it is very challenging to select appropriate projects within budget to maximize user benefit. Projects selection to update the Dhaka city network through the TNDP-PSO might ensure cost benefit that is very important for developing country Bangladesh.

B. TNDP-PSO Solutions for Dhaka City Network Update

Table I (at the end of the paper) shows the approximate parameter values of the arcs of the city network for travel time function for Eq. (5). The values of α is given in hours and β in hours per thousand vehicles. On the other hand origin/ destination (OD) wise traffic demand data are taken from a Dhaka Bus Map. The OD demands of first 30 nodes from total 77 locations are shown in Table II (at the end of the paper). The demand is given thousand vehicles per hour.

To update the network, Table III shows 20 project arcs ($|\bar{A}| = 20$), of which 10 projects are improvement on existing arcs, and other 10 are new arcs. In the remark column each project is mentioned whether the project is to develop a new read or update existing road. As an example, project P7 is to construct a new road between node 18 (Kakoli) and 75 (Mirpur 14). For a project parameters of the travel time functions (i.e., values α and β) and the construction cost (in units of thousand Taka BDT). There are $2^{20}(= 1048576)$ alternative options for the 20 projects. The projects are not real but considered overlooking the OD traffic demand

TABLE III. PARAMETER OF TRAVEL TIME FUNCTIONS AND THE CONSTRUCTION COSTS OF THE PROJECT ARCS.

(α is given in hours, and β in hours per thousand vehicles)

Project	Arc	Parameter		Const. Cost	Remark
		α	β		
P1	(20, 13), (13, 20)	0.02	0.00000237	1650	New
P2	(59, 64), (64, 59)	0.04	0.00000356	1250	New
P3	(50, 54), (54, 50)	0.02	0.00000268	1000	New
P4	(51, 72), (72, 51)	0.05	0.00000435	600	Update
P5	(18, 19), (19, 18)	0.04	0.00000524	900	Update
P6	(72, 19), (19, 72)	0.03	0.00000032	550	Update
P7	(18, 75), (75, 18)	0.02	0.00000224	1150	New
P8	(70, 75), (75, 70)	0.03	0.00000041	1800	New
P9	(6, 18), (18, 6)	0.06	0.00000203	1050	Update
P10	(49, 50), (50, 49)	0.03	0.00000316	400	Update
P11	(45, 46), (46, 45)	0.06	0.00000716	250	Update
P12	(51, 71), (71, 51)	0.02	0.00000016	1550	New
P13	(5, 6), (6, 5)	0.05	0.00000516	250	Update
P14	(19, 71), (71, 19)	0.03	0.00000216	1200	New
P15	(10, 11), (11, 10)	0.08	0.00000216	550	Update
P16	(73, 76), (76, 73)	0.02	0.00000116	1450	New
P17	(6, 73), (73, 6)	0.05	0.00000516	2800	New
P18	(20, 21), (21, 20)	0.03	0.00000216	850	Update
P19	(25, 17), (17, 25)	0.02	0.00000016	1550	New
P20	(59, 60), (60, 59)	0.02	0.00000116	950	Update

and map of Fig. 2. TNDP-PSO will seek appropriate projects within budget to ensure maximum traffic mobility.

The TNDP-PSO algorithm for Dhaka city network update was implemented on Visual C++ of Visual Studio 2013. The experiments have been done on a PC (Intel Core i5-3470 CPU @ 3.20 GHz CPU, 4GB RAM) with Windows 7 OS. In the experiment, number of particles in the population size was 20; the value of inertia factor ω was varied from 2.4 to 0.4; and the value of both the learning factors (i.e., c_1 and c_2) was considred as 2. The selected parameters are not optimal values, but considered for simplicity as well as for fairness in observation.

Table IV presents TNDP-PSO solutions to consider projects from Table III for different budget amounts. The budget utilized is the total cost to implement the selected projects. Optimal budget utilization is a key issue in TNDP and it is seen that the solutions provided the algorithm is always greater than 95% of budget utilization. Moreover, for larger budget amount more projects selected to implement. More significantly, the method always considered the important project(s) for optimal increase of traffic mobility. The project P3, new link between nodes 50 (i.e., Kalabagan) and 50 (i.e., Dhanmondi) is considered in all the solutions that is an important link to increase alternate road option in Section C and hence increase traffic mobility. The project P7 is also an important new link between nodes 18 (i.e., Kakoli) and 75 (i.e., Mirpur) to establish traffic flow option between node sections A and B because at present (as of Fig. 2) traffic moves between the node sections through usually using node 72 of node section C. Since the P7 is relatively larger amount project the algorithm considered it in the cases for budget more than 6000 (i.e., Solutions 4 to 8). The method also equally considers update link projects as like new projects to increase traffic flow. Projects P4, P5 and P6 seem to be the most important links in the existing network (Fig. 2) and therefore the projects are alternatively selected in most of the solutions. Finally the TNDP-PSO is found an effective tool to update Dhaka city network.

TABLE IV. TNDP_PSO SOLUTIONS FOR FOR VARIOUS BUDGET LEVELS FOR DHAKA CITY NETWORK UPDATE.

Soln.	Budget	Budget Utilized	Budget Utilization	Selected Projects
1	2500	2400	96.00%	P3, P4, P6, P11
2	3150	3050	96.82%	P3, P4, P5, P11, P15
3	5000	4900	98.00%	P3, P5, P9, P10, P12
4	6075	5950	97.94%	P1, P3, P7, P9, P11, P18
5	7650	7600	99.35%	P2, P3, P7, P8, P11, P14, P20
6	9000	9000	100.00%	P1, P2, P3, P7, P8, P9, P11, P18
7	11000	10800	98.18%	P1, P2, P3, P5, P7, P8, P10, P11, P14, P16, P20
8	13000	12550	96.54%	P1, P2, P3, P5, P7, P8, P9, P10, P11, P13, P14, P16, P19, P20

IV. CONCLUSIONS

Transportation network is a key issue for urban areas and its development is always an important issue to fulfill the growing demand. However, network development is always bears a challenging issue to modify appropriate existing road(s) and/or optimal construction of new road(s) considering some constraints to maximize user benefit through optimal utilization of resources. In this regard, transportation network design problem (TNDP) has been emerged as a challenging optimization task for optimal selection of several projects from various alternative proposed projects with estimated costs. Among various approaches to solve TNDP, PSO based TNDP-PSO is the recent one and is shown to outperform others. Most of the existing methods including PSO considered to solve small size network such as Sioux Falls [2, 3].

This study investigated TNDP-PSO for network development of the megacity Dhaka. Considered road network consists of 77 nodes and 192 arcs; and is almost 10 times larger than network of Sioux Falls. With roughly estimated network parameters of the links (i.e., α and β), experiments were conducted for 20 imaginary projects those seem to be important. The experimental results reveal that TNDP-PSO is able to select appropriate projects to maximize user benefit through optimal (or near optimal) utilization of a given budget.

A number of potential works remain after this study. Recently, govt. already developed a big project to connect node 6 (i.e., Kuril) and node 75 (i.e., Mirpur 14). On the other hand, the algorithm provided alternate solutions of it with project P9 (updating arc between nodes 6 and 9) and P7 (new arc between nodes 18 and 75). This indicates experiments with practical data might give more feasible result. Moreover, TNDP is always incur high computation cost [3], the result presented in this study only for few trials as well as interaction. More optimal result may get with more trials with different parameter settings using high configuration computer. As an alternate, the technique to solve the problem might be interesting to solve it by dividing the network into smaller parts.

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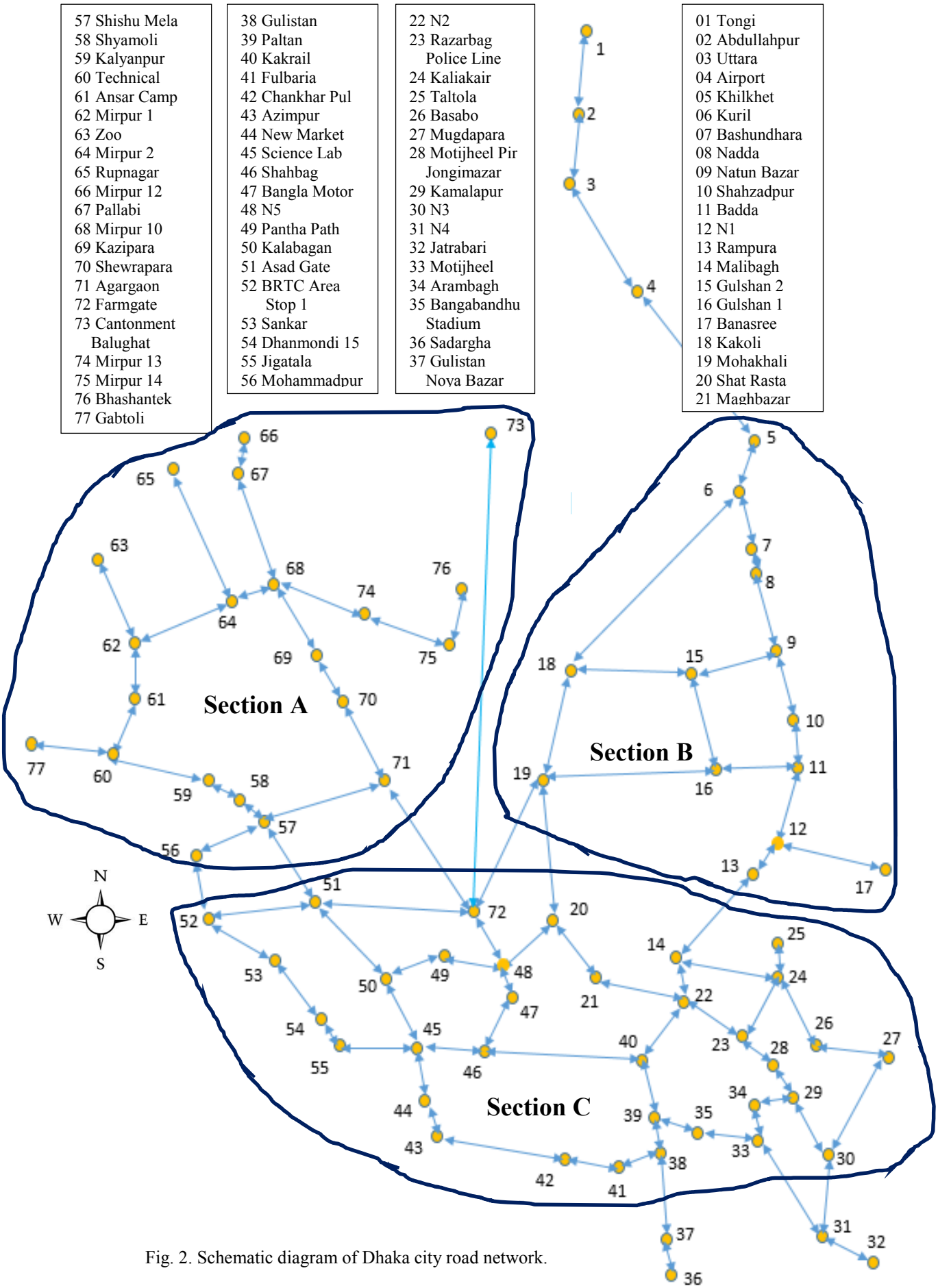

Fig. 2. Schematic diagram of Dhaka city road network.

TABLE I. PARAMETERS OF THE TRAVEL TIME FUNCTIONS FOR THE NETWORK OF DHAKA CITY SHOWN IN FIGURE 2.

(α is given in hours, and β in hours per thousand vehicles)

Arcs	Parameters		Arcs	Parameters		Arcs	Parameters		Arcs	Parameters	
	α	β		α	β		α	β		α	β
(1, 2), (2, 1)	0.02	0.00002004	(14, 24), (24, 14)	0.06	0.00000732	(46, 45), (45, 46)	0.10	0.00001060	(59, 58), (58, 59)	0.03	0.00001557
(2, 3), (3, 2)	0.04	0.00002875	(26, 24), (24, 26)	0.06	0.00003374	(50, 45), (45, 50)	0.08	0.00002018	(58, 59), (59, 58)	0.09	0.00001032
(3, 4), (4, 3)	0.07	0.00001707	(24, 26), (26, 24)	0.08	0.00000387	(55, 45), (45, 55)	0.04	0.00002707	(60, 59), (59, 60)	0.04	0.00002734
(4, 5), (5, 4)	0.05	0.00003185	(26, 27), (27, 26)	0.07	0.00001371	(39, 46), (46, 39)	0.09	0.00001415	(59, 60), (60, 59)	0.05	0.0000072
(5, 6), (6, 5)	0.09	0.00002655	(23, 28), (28, 23)	0.08	0.00002166	(40, 46), (46, 40)	0.06	0.00003165	(61, 60), (60, 61)	0.07	0.00002149
(6, 7), (7, 6)	0.03	0.00003054	(24, 28), (28, 24)	0.07	0.00001286	(45, 46), (46, 45)	0.02	0.00003536	(77, 60), (60, 77)	0.06	0.00001659
(7, 8), (8, 7)	0.08	0.00001827	(34, 28), (28, 34)	0.03	0.00000802	(47, 46), (46, 47)	0.07	0.00000181	(60, 61), (61, 60)	0.07	0.00002257
(8, 9), (9, 8)	0.04	0.00000055	(26, 32), (32, 26)	0.05	0.00002855	(21, 47), (47, 21)	0.06	0.00000692	(62, 61), (61, 62)	0.09	0.00002947
(9, 10), (10, 9)	0.03	0.00001297	(27, 32), (32, 27)	0.06	0.00003519	(46, 47), (47, 46)	0.09	0.00002229	(61, 62), (62, 61)	0.07	0.00002563
(10, 11), (11, 10)	0.03	0.00000591	(28, 32), (32, 28)	0.08	0.0000123	(72, 47), (47, 72)	0.07	0.00000705	(63, 62), (62, 63)	0.07	0.00001335
(12, 11), (11, 12)	0.10	0.00001586	(33, 32), (32, 33)	0.03	0.00002337	(20, 49), (49, 20)	0.09	0.00000440	(64, 62), (62, 64)	0.03	0.00002624
(11, 12), (12, 11)	0.03	0.00000019	(35, 32), (32, 35)	0.06	0.00000228	(47, 49), (49, 47)	0.03	0.00002642	(62, 64), (64, 62)	0.06	0.00003218
(13, 12), (12, 13)	0.02	0.00001345	(32, 33), (33, 32)	0.08	0.00001796	(45, 50), (50, 45)	0.05	0.00003346	(65, 64), (64, 65)	0.04	0.00000673
(17, 12), (12, 17)	0.06	0.00002031	(34, 33), (33, 34)	0.03	0.00003376	(49, 50), (50, 49)	0.04	0.00001197	(68, 64), (64, 68)	0.07	0.00002484
(12, 13), (13, 12)	0.07	0.00002159	(35, 33), (33, 35)	0.03	0.00003218	(51, 50), (50, 51)	0.03	0.00002607	(66, 67), (67, 66)	0.07	0.0000125
(24, 13), (13, 24)	0.03	0.00002358	(39, 33), (33, 39)	0.08	0.00001079	(50, 51), (51, 50)	0.09	0.00002518	(64, 68), (68, 64)	0.06	0.00000288
(13, 14), (14, 13)	0.06	0.00001253	(27, 34), (34, 27)	0.05	0.00000252	(57, 51), (51, 57)	0.07	0.00002657	(67, 68), (68, 67)	0.08	0.00002177
(22, 14), (14, 22)	0.02	0.00002161	(28, 34), (34, 28)	0.1	0.00002429	(58, 51), (51, 58)	0.04	0.00000515	(69, 68), (68, 69)	0.07	0.00002458
(16, 15), (15, 16)	0.08	0.00002854	(35, 34), (34, 35)	0.03	0.00003119	(72, 51), (51, 72)	0.02	0.00000219	(74, 68), (68, 74)	0.08	0.00000532
(11, 16), (16, 11)	0.06	0.00001075	(33, 35), (35, 33)	0.09	0.0000207	(51, 52), (52, 51)	0.08	0.00003031	(68, 69), (69, 68)	0.07	0.00003085
(15, 16), (16, 15)	0.09	0.00002584	(34, 35), (35, 34)	0.04	0.00000634	(53, 52), (52, 53)	0.04	0.00000413	(70, 69), (69, 70)	0.09	0.00002186
(5, 18), (18, 5)	0.10	0.00003291	(39, 35), (35, 39)	0.09	0.00001691	(56, 52), (52, 56)	0.06	0.00000053	(69, 70), (70, 69)	0.08	0.00000156
(6, 18), (18, 6)	0.06	0.00000508	(39, 37), (37, 39)	0.03	0.00001792	(52, 53), (53, 52)	0.03	0.00001617	(71, 70), (70, 71)	0.07	0.00003472
(15, 18), (18, 15)	0.06	0.00000838	(33, 39), (39, 33)	0.08	0.00001443	(54, 53), (53, 54)	0.08	0.00002440	(19, 71), (71, 19)	0.05	0.00002024
(16, 19), (19, 16)	0.09	0.00000747	(34, 39), (39, 34)	0.04	0.00002023	(53, 54), (54, 53)	0.06	0.00000265	(70, 71), (71, 70)	0.04	0.0000062
(18, 19), (19, 18)	0.08	0.00003000	(35, 39), (39, 35)	0.07	0.00002688	(55, 54), (54, 55)	0.05	0.00000719	(19, 72), (72, 19)	0.03	0.0000309
(19, 20), (20, 19)	0.10	0.00003554	(40, 39), (39, 40)	0.08	0.00001691	(45, 55), (55, 45)	0.08	0.00001034	(47, 72), (72, 47)	0.09	0.00002647
(20, 21), (21, 20)	0.07	0.00001396	(46, 39), (39, 46)	0.03	0.00001309	(54, 55), (55, 54)	0.05	0.00000828	(51, 72), (72, 51)	0.03	0.00001164
(22, 21), (21, 22)	0.04	0.00001058	(47, 39), (39, 47)	0.09	0.00000127	(52, 56), (56, 52)	0.07	0.00001894	(71, 72), (72, 71)	0.03	0.00000274
(47, 21), (21, 47)	0.09	0.00000086	(22, 40), (40, 22)	0.06	0.00002358	(58, 56), (56, 58)	0.07	0.00000571	(73, 72), (72, 73)	0.07	0.00002915
(72, 21), (21, 72)	0.05	0.00000331	(39, 41), (41, 39)	0.05	0.00000374	(19, 57), (57, 19)	0.06	0.00003424	(68, 74), (74, 68)	0.06	0.00001595
(13, 22), (22, 13)	0.07	0.00000202	(42, 41), (41, 42)	0.1	0.00003276	(45, 57), (57, 45)	0.08	0.00003288	(75, 74), (74, 75)	0.05	0.00001063
(14, 22), (22, 14)	0.02	0.00003266	(43, 42), (42, 43)	0.06	0.00001231	(51, 57), (57, 51)	0.04	0.00001196	(18, 75), (75, 18)	0.06	0.00001783
(21, 22), (22, 21)	0.04	0.00000972	(44, 43), (43, 44)	0.06	0.00001334	(58, 57), (57, 58)	0.03	0.00003538	(74, 75), (75, 74)	0.03	0.0000115
(22, 23), (23, 22)	0.07	0.00002458	(45, 44), (44, 45)	0.09	0.00001128	(71, 57), (57, 71)	0.06	0.00003548	(76, 75), (75, 76)	0.08	0.00001117
(13, 24), (24, 13)	0.09	0.00002583	(50, 44), (44, 50)	0.06	0.00000968	(57, 58), (58, 57)	0.03	0.00002223	(60, 77), (77, 60)	0.09	0.0000341

TABLE II. ORIGIN/ DESTINATION WISE TRAFFIC DEMAND IN THOUSAND VEHICLES PER HOUR FOR FIRST 30 NODES OUT OF 77 NODES.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	0	15	15	15	15	15	5	5	5	5	5	0	5	5	0	0	0	10	10	5	4	0	2	2	0	3	0	2	0	0
2	15	0	21	21	21	19	6	6	6	6	6	0	6	6	0	0	0	15	15	5	4	0	2	2	0	3	0	2	0	0
3	15	21	0	21	21	19	6	6	6	6	6	0	6	6	0	0	0	15	15	5	4	0	2	2	0	3	0	2	0	0
4	15	21	21	0	22	20	6	6	6	6	6	0	6	6	0	0	0	16	16	5	4	0	2	2	0	3	0	2	0	0
5	15	21	21	22	0	20	6	6	6	6	6	0	6	6	0	0	0	16	16	5	4	0	2	2	0	3	0	2	0	0
6	15	19	19	20	20	0	8	8	8	8	8	0	6	6	0	2	0	14	16	7	4	0	2	2	0	3	0	2	0	0
7	5	6	6	6	6	8	0	8	8	8	8	0	6	6	0	2	0	0	2	2	0	0	0	0	0	3	0	0	0	0
8	5	6	6	6	6	8	8	0	9	9	9	0	7	7	0	2	0	0	2	2	0	0	0	0	0	3	0	0	0	0
9	5	6	6	6	6	8	8	9	0	9	9	0	7	7	0	2	0	0	2	2	0	0	0	0	0	3	0	0	0	0
10	5	6	6	6	6	8	8	9	9	0	9	0	7	7	0	2	0	0	2	2	0	0	0	0	0	3	0	0	0	0
11	5	6	6	6	6	8	8	9	9	9	0	0	9	7	0	6	1	0	6	2	0	0	0	0	0	3	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	5	6	6	6	6	6	6	7	7	7	9	0	0	7	0	2	3	0	2	0	1	0	0	0	0	4	1	0	0	0
14	5	6	6	6	6	6	6	7	7	7	7	0	7	0	0	0	0	0	0	0	1	0	0	0	0	4	1	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	2	3	1	2	0	0	0	0	0	0	2	0	0
16	0	0	0	0	0	2	2	2	2	2	6	0	2	0	3	0	1	2	9	3	2	0	0	0	0	0	0	2	0	0
17	0	0	0	0	0	0	0	0	0	0	1	0	3	0	0	1	0	0	1	0	1	0	0	0	0	1	1	0	0	0
18	10	15	15	16	16	14	0	0	0	0	0	0	0	0	2	2	0	0	18	6	6	0	2	2	0	0	0	4	0	0
19	10	15	15	16	16	16	2	2	2	2	6	0	2	0	3	9	1	18	0	8	6	0	2	2	0	0	0	4	0	0
20	5	5	5	5	5	7	2	2	2	2	2	0	0	0	1	3	0	6	8	0	5	0	2	2	0	0	0	3	0	0
21	4	4	4	4	4	4	0	0	0	0	0	0	1	1	2	2	1	6	6	5	0	0	3	2	0	1	1	4	0	0
22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	2	2	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	2	2	2	3	0	0	0	0	0	0	2	0	0
24	2	2	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	2	2	2	2	0	0	0	0	0	0	0	0	0
25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
26	3	3	3	3	3	3	3	3	3	3	3	0	4	4	0	0	1	0	0	0	1	0	0	0	1	0	4	1	0	0
27	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0	1	0	0	0	1	4	0	0	0	0
28	2	2	2	2	2	2	0	0	0	0	0	0	0	0	2	2	0	4	4	3	4	0	2	0	0	1	0	0	0	0
29	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0